

Multiple Choice (10 marks)

1. _____ is also a part of darknet that is employed for transferring files anonymously.
 - (a) Freenet
 - (b) ARPANET
 - (c) Stuxnet
 - (d) Internet
2. Why apps developed in languages like C, C++ is prone to Buffer-overflow?
 - (a) No string boundary checks in predefined functions
 - (b) No storage check in the external memory
 - (c) No processing power check
 - (d) No database check
3. Can a stream cipher have perfect secrecy?
 - (a) Yes, if the PRG is really "secure"
 - (b) No, there are no ciphers with perfect secrecy
 - (c) Yes, every cipher has perfect secrecy
 - (d) No, since the key is shorter than the message
4. SHA-1 has a message digest of
 - (a) 160 bits
 - (b) 512 bits
 - (c) 628 bits
 - (d) 820 bits
5. The Message confidentiality or privacy means that the sender and the receiver expect
 - (a) Integrity
 - (b) Confidentiality
 - (c) Authentication
 - (d) Nonrepudiation

True / False (10 marks)

Answer True or False

6. True or False: Encapsulating Security Payload (ESP) is a component of the Secure IP Protocol used to provide encryption and authentication in IP communications.
7. True or False: A buffer is a sequential segment of memory allocated to store data, such as character strings or arrays of integers.
8. True or False: WPA uses TKIP as a security protocol to enhance encryption and ensure data integrity in wireless networks.
9. True or False: SQL queries based on user input are highly vulnerable to injection attacks due to improper handling of user-supplied data.
10. True or False: ECB, CFB, and CBC are all examples of valid block cipher operating modes.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find sum and average of first n natural numbers.
12. Write a function to remove consecutive duplicates of a given list.

End of Paper